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MMRV vaccine safety: A comprehensive analysis of adverse events recorded among pediatric populations

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ABSTRACT

Measles, mumps, rubella, and varicella (MMRV) vaccines are widely used in global immunization programs, but safety data on their administration in children remain limited. We analyzed adverse events following immunization(AEFIs) associated with the MMRV vaccine in children under seven y of age, as reported to the US Vaccine Adverse Event Reporting System (VAERS) from 2005 to 2025. Disproportionality analyses were applied, with multiple testing correction using a 5% false discovery rate. Subgroup assessments focused on children with asthma, epilepsy, and chronic kidney disease. The results show that among the 13,724 reports, 94.1% were non-serious cases, and the signal of Conjunctival Disorder had the highest reporting rate (ROR = 61.13, 95%confidence interval[CI]: 6.36-587.74). The risk of seizures in children with epilepsy significantly increased (ROR = 10.27, 95% CI: 6.57-16.05), with 63.2% occurring within 48 h after vaccination; a 6.81-fold increase in ROR was observed in the risk of wheezing in children with asthma, and no significant risk increment was observed in the chronic kidney disease group. Susceptibility to risks varied across subgroups. Weibull distribution analysis indicated that AEFIs following MMRV exhibited early-onset characteristics. This study provides a comprehensive understanding of MMRV vaccine safety and highlights the need for targeted protective measures in children with epilepsy and asthma to reduce vaccine hesitancy and improve vaccination coverage.

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Introduction

Measles, mumps, rubella and varicella are highly contagious childhood diseases that can be prevented by vaccination. Global implementation and safety monitoring of the combined measles, mumps, rubella and varicella vaccines (MMRV) are directly linked to the effectiveness of public health policies. Since the inclusion of MMRV in the US vaccination program in 2005, the protection rate against these diseases has exceeded 95%,[1] and in some regions, local elimination has been achieved.[2] However, in children with underlying chronic diseases, immune response mechanisms may differ from those of healthy children, and their physiological conditions may affect the immunogenicity of the vaccine, thereby altering the risk characteristics of adverse events following immunization (AEFIs). Therefore, a systematic assessment based on real-world data is urgently needed.[3]

Previous studies leveraging VAERS or the Vaccine Safety Datalink (VSD), such as Jacobsen et al.[4] and Klein et al.[5] have provided valuable insights into specific AEFIs like fever and febrile seizures following MMRV vaccination. However, these studies often focused on the general pediatric population or specific AEFIs, were frequently limited by a single signal detection method or a short observation period and paid less attention to children with chronic conditions who are often excluded from clinical trials. In the vaccine phase III trial, the proportion of children with

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epilepsy was less than 0.5%, making it impossible to assess the neurological risks.[6] At the same time, the duration between receiving the first dose at 12 to 15 months and the second dose at 4 to 6 y precisely covers the high-risk diagnosis window of chronic diseases in children.[7] These limitations have exacerbated the issue of vaccine hesitancy among this population.[8] Our study builds upon this foundational work by conducting a comprehensive, multi-method disproportionality analysis on a substantially longer timeline of VAERS data (2005–2025). More importantly, it extends previous research by specifically investigating the vaccination risks in key pediatric chronic disease subgroups (asthma, epilepsy, and chronic kidney disease)–a population with significant real-world data gaps. Additionally, we innovatively employ Weibull distribution analysis to characterize the temporal onset patterns of AEFIs, providing new insights into the risk evolution over time post-vaccination.

The goal of this study was to address these issues by analyzing data from the US Vaccine Adverse Event Reporting System (VAERS): (1) to use disproportionality analysis to quantify the AEFI risks of the MMRV vaccine in real-world pharmacovigilance; (2) to identify specific differences and increases in AEFIs among three groups of children with underlying diseases such as asthma, epilepsy, and chronic kidney disease; and (3) to compare AEFI the differences among subgroups, and assess trends in AEFI onset time. The rationale of this study was to provide insights for the development of management strategies and prevention programs targeting willingness to receive MMRV vaccination and the coverage of immunization programs, representing an important and urgent public health concern.

Methods

Data source

Reports of AEs associated with MMRV vaccine from 2005 to 2025 were extracted from the VAERS database[9] (Accessed 10th August 2025). Prior to analysis, potential duplicate reports were identified and removed based on the unique VAERS ID, ensuring each case contributed only once to the analysis. The VAERS database is used to identify potential safety issues related to authorized vaccines and does not record sensitive personal information (e.g., name or ID number) of individuals. It uses the Medical Dictionary for Regulatory Activities (MedDRA) preferred terms (PT) to code AEFIs. In MedDRA, although a single PT may correspond to multiple system organ classes (SOCs), it is assigned to only one primary SOC which was used to classify the PTs in this study The false discovery rate (FDR) was controlled at 5% using the Benjamini-Hochberg procedure. As a passive surveillance system, VAERS is subject to underreporting and reporting biases, which are acknowledged as inherent limitations of the data source.

Inclusion criteria

(1) Vaccine types: MMRV, including two categories in the database–MMRV (PROQUAD) and MMRV (NO BRAND NAME); (2) Age criterion: ≤7 y. (3) Subgroups with underlying diseases: children with chronic diseases who were clinically eligible for MMRV vaccination, including epilepsy_(the CUR_ILL or HISTORY field contained “epilepsy,”seizure,” or “convulsion”); asthma (the CUR_ILL or HISTORY field contained “asthma”); and chronic kidney disease (the CUR_ILL or HISTORY field contained “renal,” “proteinuria,” or “kidney”). After the initial screening, cases were confirmed for inclusion through manual verification; (4) Control group vaccines: the measles, mumps, and rubella triple vaccine (MMR) and the varicella vaccine. MMR included five categories: MMR (MMR I), MMR (MMR II), MMR (NO BRAND NAME), MMR (PRIORIX), and MMR (VIRIVAC). The varicella vaccine included VARICELLA (VARIVAX) and VARICELLA (VARILRIX). The age range was consistent with that of the MMRV vaccine group. After the initial screening, all potentially eligible cases underwent a structured manual verification process to confirm the presence of the preexisting chronic condition. This verification was independently performed by two trained researchers. Any discrepancies between the two reviewers were resolved through consensus discussion, and if necessary, by adjudication from a senior investigator. It ensured the accurate and reproducible identification of cases for the chronic disease subgroups.

Disproportionality analysis

Four disproportionality measures were used to identify possible associations between MMRV and AEFIs: the reporting odds ratio (ROR), proportional reporting ratio (PRR), Bayesian confidence propagation neural network (BCPNN),[10] and multi-item gamma Poisson shrinker (MGPS) (Table S1). This analysis calculated the ratio of the incidence rate of the target adverse event in MMRV group to that in all other vaccines. When the frequency of the target adverse event caused by the target vaccine was higher than that of all other vaccines, it indicated a significant safety signal (Table S2). We have established a strict screening procedure, which stipulates that when all four methods yield positive results, a positive signal is confirmed. To determine the presence or absence of statistical differences in AEFI incidence, hierarchical logistic regression analysis was performed, grouping data by age, sex, vaccine dose, and vaccine type.

Designated medical event (DME) list screening

A comparative analysis was conducted using the European Medicines Agency (EMA) 's DME list to evaluate serious and specific safety events related to MMRV and to identify valuable positive signals.[11] The DME list was developed based on the preferred terms of MedDRA and serves as a key pharmacovigilance tool to prioritize the review of serious AEFIs. This ensured that reports of serious AEFIs were not overlooked, even if they did not reach statistical significance.

Weibull distribution analysis

The Weibull distribution test was applied to evaluate trends in AEFI incidence over time using scale (α) and shape (β) parameters. A shape parameter $\beta < 1$ with a 95% confidence interval (CI) < 1 indicated a decreasing incidence (early failure pattern); $\beta \approx 1$ with a 95% CI including 1 reflected a constant incidence (random failure pattern); and $\beta > 1$ with a 95% CI excluding 1 suggested an increasing incidence (wear-out failure pattern).

Sensitivity analysis

If a signal is marked as a valid signal by the three algorithms ROR, BCPNN and MGPS, it will still be regarded as a positive signal. Through this more lenient threshold limitation, we evaluate the reliability of our research results. The focus is on whether the newly identified signals and the key signals under the new standard still exist, providing a more solid basis for public health decisions.

Results

Descriptive overview

From the approval of the MMRV vaccine in September 2005 to May 2025, 269,077 AEFIs were reported, of which 13,724 cases were included in this study. The annual number of reports peaked in 2007, with 1,405 cases and subsequently declined in the following y. However, the number has remained relatively stable at an average of 900 cases per year since 2013 (with a transient decrease during the COVID-19 pandemic period) and reached a peak in 2017 (1,234 cases) (Figure S1). A total of 810 serious cases (5.9%,) and 31 deaths were reported. The reporting age group was concentrated within the recommended vaccination age range, and the male-to-female ratio was approximately equal, with 11% of cases not reported. Among the reports, 4,955 (36.1%) involved the first dose, 3,350 (24.4%) involved the second dose, and the Merck & Co. ProQuad® vaccine accounted for 96.7% of cases (Table 1). A flowchart of this study is shown in Figure 1.

Disproportionality signals

Building upon these descriptive findings, we further conducted disproportional analysis to detect potential safety signals. A total of 2,183 PT signals were obtained. By analyzing four disproportional algorithms (ROR, PRR, MGPS and BCPNN) and excluding signals clearly unrelated to the

Table 1. Characteristics of VAERS reports of MMRV vaccines administered to children aged less than 7 y, 2005–2025.

Characteristic	<1 y	1–1.5 y	1.5–4 y	4–7 y	≥7 y	Overall
	(N = 214)	(N = 4013)	(N = 920)	(N = 8436)	(N = 141)	(N = 13724)
Sex						
Famale	74 (34.6%)	1691 (42.1%)	404 (43.9%)	3660 (43.4%)	67 (47.5%)	5896 (43.0%)
Male	78 (36.4%)	1874 (46.7%)	389 (42.3%)	3914 (46.4%)	60 (42.6%)	6315 (46.0%)
Unknown	62 (29.0%)	448 (11.2%)	127 (13.8%)	862 (10.2%)	14 (9.9%)	1513 (11.0%)
Serious cases*						
N	191 (89.3%, 95% CI: 84.3%–93.1%)	3578 (89.2%, 95% CI: 88.2%–90.1%)	827 (89.9%, 95% CI: 87.8%–91.8%)	8181 (97.0%, 95% CI: 96.6%–97.3%)	137 (97.2%, 95% CI: 92.9%–99.2%)	12914 (94.1%, 95% CI: 93.7%–94.5%)
Y	23 (10.7%, 95% CI: 6.9%–15.7%)	435 (10.8%, 95% CI: 9.9%–11.8%)	93 (10.1%, 95% CI: 8.2%–12.2%)	255 (3.0%, 95% CI: 2.7%–3.4%)	4 (2.8%, 95% CI: 0.8%–7.1%)	810 (5.9%, 95% CI: 5.5%–6.3%)
Time to onset						
Mean (SD)	24.0 (150)	26.3 (206)	39.8 (396)	11.8 (123)	2.20 (7.58)	17.9 (180)
Vaccine dose						
1	83 (38.8%)	2316 (57.7%)	443 (48.2%)	2066 (24.5%)	47 (33.3%)	4955 (36.1%)
2	9 (4.2%)	91 (2.3%)	88 (9.6%)	3119 (37.0%)	43 (30.5%)	3350 (24.4%)
>2	1 (0.5%)	15 (0.4%)	6 (0.6%)	186 (0.2%)	5 (3.5%)	213 (1.6%)
Unknown	121 (56.5%)	1591 (39.6%)	383 (41.6%)	3065 (36.3%)	46 (32.6%)	5206 (37.9%)
Vaccine name						
MMRV (No Brand Name)	18 (8.5%)	213 (5.3%)	46 (5.0%)	172 (2.0%)	3 (2.1%)	452 (3.3%)
MMRV (ProQuad)	195 (91.5%)	3800 (94.7%)	874 (95.0%)	8264 (98.0%)	138 (97.9%)	13272 (96.7%)
Data year						
2005	0	6	0	0	0	6
2006	8	284	23	139	2	456
2007	16	526	93	763	7	1405
2008	7	115	25	120	1	268
2009	1	47	8	20	0	76
2010	0	55	7	176	0	238
2011	3	47	18	185	1	254
2012	1	30	7	38	0	76
2013	6	165	40	485	3	699
2014	10	175	52	560	8	805
2015	9	188	62	552	4	815
2016	11	212	56	653	3	935
2017	17	253	59	898	7	1234
2018	13	221	57	647	9	947
2019	21	258	74	612	7	972
2020	13	213	38	444	6	714
2021	13	224	44	382	12	675
2022	25	297	84	480	14	900
2023	15	279	64	503	26	887
2024	18	319	86	577	21	1021
2025	7	99	23	202	10	341

SD, Standard Deviation. *Serious cases including five clinical outcomes records in the VAERS database: death, life-threatening events, hospitalization, prolonged hospitalization, and disability.

adverse reactions, 40 positive PT signals across 10 SOCs were identified (Figure 2(A)). The three most frequently reported symptoms were Injection site erythema, Injection site swelling, and Injection site warmth. At the SOC level, they all belonged to General disorders and administration site conditions. Conjunctival disorder (ROR = 61.13, 95% CI: 6.36-587.74), Cutaneous vasculitis (ROR = 20.38, 95% CI: 4.11-100.97), and Oppositional defiant disorder (ROR = 14.56, 95% CI: 4.62-45.87) exhibited the highest ROR values. At the SOC level, these were classified under Eye disorders, Skin and subcutaneous tissue disorders, and Psychiatric disorders. All positive signals discussed herein remained statistically significant after controlling the false discovery rate (FDR < 0.05). We also analyzed temporal variations in AEFI onset time (Figure 2(B,C)). The onset intervals were divided into 10 groups (<1 d, 17 d, 8–15 d, 16–30 d, 31–60 d, 61–90 d, 91–120 d, 121–180 d, 181–360 d, >360 d). Results showed that 96.8% of AEFIs occurred within 30 d, and 46.0% occurred within 24 hours. Weibull distribution analysis indicated that the shape parameter (β) was less than 1(β = 0.49), with the 95% CI also less than 1 (95%CI: 0.48–0.50), suggesting an early failure pattern, and indicating that the incidence of AEFIs decreased over time (Table 2).

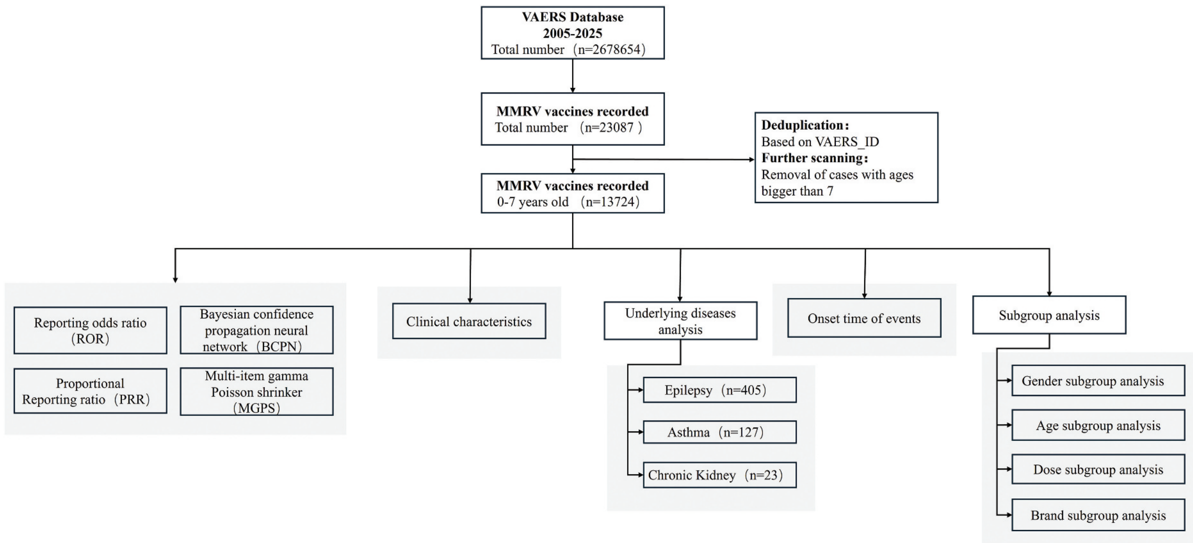

Figure 1. The process outlines the search and analysis of MMRV-related AEFIs from VAERS.

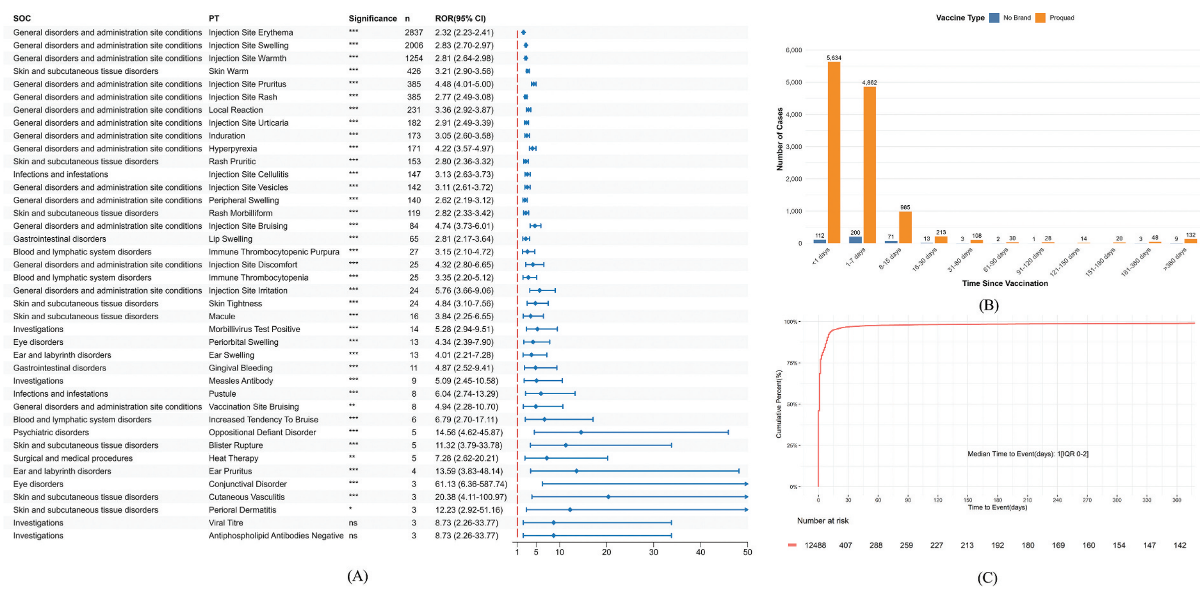

Figure 2. (A) Forest plot of PT cases for MMRV vaccines. The blue arrows indicate that the lower boundary of the 95% confidence interval for the ROR is higher than 50. The red dashed line indicates the reference line of ROR = 1, which represents no statistical association. (B) Onset time distribution of AEFIs for MMRV vaccines. (C) Curve plot of AEFIs onset time for MMRV vaccines. SOC, system organ class; PT, preferred term; ROR, reporting odds ratio; CI, confidence interval. Significance levels: *$p < .05$, **$p < .01$, ***$p < .001$ (adjusted P-value).

Table 2. Weibull distribution of MMRV AEFIs.

	Time to onset (d)		Weibull distribution		
Case n	Median (IQR)	Min-Max	Scale parameter (α 95% CI)	Shape parameter (β 95% CI)	Failure type
12,488	1(0–2)	0–10089	7.19(6.82–7.56)	0.49(0.48–0.50)	Early failure

n, number of reports that include available time-to-onset data; IQR, interquartile range; CI, confidence interval; α, Weibull scale parameters; β, Weibull shape parameters.

DME list comparison

Comparison of the 40 positive PT signals with the DME list identified one key signal: Immune thrombocytopenia (n = 25, ROR = 3.35, 95% CI: 2.2-5.12). This signal was classified at the SOC level under Blood and lymphatic system disorders and warrants special attention.

Risks for children with chronic underlying diseases

Three common chronic diseases in children, namely asthma, epilepsy and chronic kidney disease, were selected from the respiratory, nervous and urinary systems for MMRV vaccination risk analysis (Figure 3 (A)).After screening, 127 cases with 26 positive PT signals were included in the asthma group. The top three reported symptoms were Injection site erythema, Injection site swelling, and Injection site warmth, consistent with the overall situation. The top 3 highest ROR values were Skin tightness (ROR = 18.75, 95% CI: 6.92-50.83), Injection site discomfort (ROR = 12.05, 95% CI: 3.83-37.84), and Injection site bruising (ROR = 7.71, 95% CI: 3.44-17.29). At the SOC level, these were associated with Skin and subcutaneous tissue disorders and General disorders and administration site conditions. A total of 93.3% of AEFIs occurred within 7 d of vaccination (Figure S2(A)). Comparison of PT signals between the asthma group and children without underlying chronic diseases revealed 14 unique signals in the asthma group(Figure 3(B)). Wheezing (ROR = 6.81, 95% CI: 4.37-10.61) and Respiratory tract congestion (ROR = 5.12, 95% CI: 2.29-11.45) were two respiratory system abnormal signals with increased risks. Of these, 65.4% occurred within 24 hours, requiring particular attention. Among the 12 shared signals, no statistically significant differences were observed in AEFI risk. No significant difference was recorded in AEFIs onset time between the asthma group and non-chronic diseases group ($P = .19 > .05$).

The epilepsy group included 405 cases and exhibited eight positive PT signals. The three most frequently reported were Febrile convulsion, Seizure, and Condition aggravated. At the SOC level, these were categorized under Nervous system disorders and General disorders and administration site conditions.

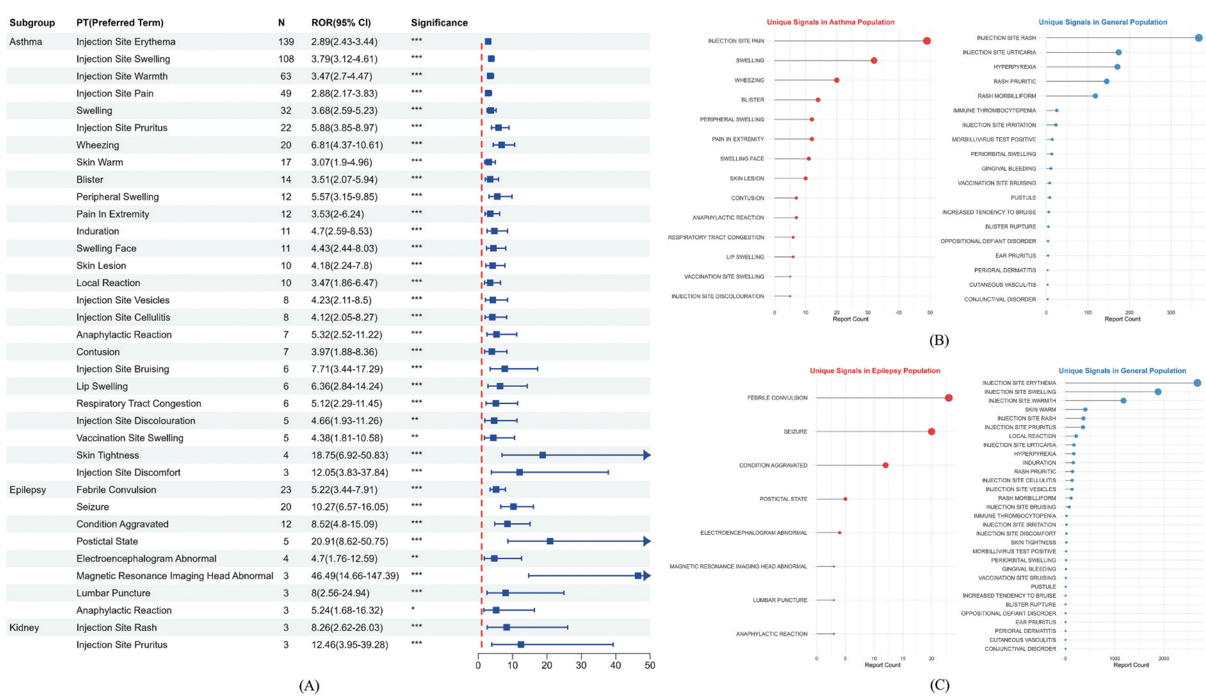

Figure 3. (A) Forest plot of positive PT signals for MMRV vaccination in children with asthma, epilepsy and chronic kidney disease. The red dashed line indicates the reference line of ROR = 1, which represents no statistical association. The blue arrows indicate that the lower boundary of the 95% confidence interval for the ROR is higher than 50. (B) Comparison of unique PT signals between the asthma group and populations with non-chronic diseases. (C) Comparison of unique PT signals between the epilepsy group and populations with non-chronic diseases. PT, preferred term; ROR, reporting odds ratio; CI, confidence interval.

The three highest ROR values were for Magnetic resonance imaging head abnormal (ROR = 46.49, 95% CI: 14.66-147.39), Postictal state (ROR = 20.91, 95% CI: 8.62-50.75), and Seizure (ROR = 10.27, 95% CI: 6.57-16.05). At the SOC level, these were classified under Investigations and Nervous system disorders. A total of 80.8% of AEFIs occurred within 7 d after vaccination (Figure S2(B)). Comparison of PT signals between the epilepsy and non-chronic groups showed that all positive signals in the epilepsy group were unique, five of which were related to brain nerve abnormalities (Figure 3(C)). Among these, 21.8% occurred within 24 h.

The chronic kidney disease group included 23 cases with two positive PT signals identified: Injection site rash and Injection site pruritus. These were grouped in the SOC category as General disorders and administration site conditions. A total of 90% of AEFIs were recorded within 7 d of vaccination (Figure S2(C)). Comparison of PT signals between the chronic kidney disease and non-chronic diseases groups revealed no specific signals. Both positive signals were common AEFIs and did not find a statistically significant difference in their risk of occurrence. ($P = .12 > .05$, $P = .11 > .05$). However, due to the small sample size, this subgroup analysis was underpowered and should be interpreted with caution.

Subgroup analysis

Age subgroup analysis

According to the recommended vaccination schedule, we divided the study population into 5 subgroups: <1 year, >1 and <1.5 y, ≥1.5 and <4 y, ≥4 and <7 y, and ≥7 y. The numbers of cases in each subgroup were 214, 4,013, 920, 8,436 and 141 respectively. The positive PT signals that met all four disproportionality algorithms in these subgroups were 4, 15, 13, 2 and 1,respectively (Figure 4(A)). We focused on AEFIs in the 1–1.5 y and 4–7 y subgroups because they correspond to the recommended vaccination window. The number of severe cases was highest in the 1–1.5 y group (10.8%). The most frequently reported PT was Hyperpyrexia, whereas the highest reporting rate was observed for Red blood cells urine positive (ROR = 13.59, 95% CI: 3.93-46.93). Two-thirds of the positive signals involved abnormalities in body fluid examinations (e.g., blood, urine). In the 4–7 y group, two non-fatal positive PT signals were identified: Injection site bruising and Vaccination site rash. No common positive signals were observed between the two age groups. Regarding the onset time of AEFIs, differences were observed among the subgroups. In the 4–7 y group, 94.4% of AEFIs occurred within 7 d after vaccination, and the onset time was earlier than that in the 1–1.5 y group (Figure 4(B)).

Sex-based subgroup analysis

The male subgroup included 6,315 cases in which 26 positive PT signals were identified. The female subgroup comprised 5,896 cases with 34 positive PT signals detected (Figure 4(C)). The three most frequently reported PTs in both male and female subgroups were Injection site erythema, Injection site swelling, and Injection site warmth. Fourteen positive signals were shared between the two groups, whereas 20 and 12 signals were unique to males and females, respectively (Figure 4(D)). In the female subgroup, the highest RORs were observed for Thrombotic thrombocytopenic purpura (ROR = 20.49, 95% CI: 4.14-101.54) and Formication (ROR = 20.49, 95% CI: 4.14-101.54), and the most frequently reported signal was Injection site cellulitis. In the male subgroup, the most reported signal was Pain in extremity, whereas the signal with the highest ROR was Blister rupture (ROR = 14.17, 95% CI: 4-50.2). No statistically significant differences were observed between male and female subgroups in the onset time of AEFIs (Figure S3(A)).

Dose-based subgroup analysis

Routine MMRV vaccination consists of two doses. The analysis for dose subgroups was based directly on the 'VAX_DOSE_SERIES' field in VAERS. Numbers of reports for the first and second doses were 4,955 and 3,350, respectively, with 35 and 39 positive signals identified (Figure 4(E)). The most reported signal for the first dose was Injection site swelling, and the one with the highest ROR ratio was Penile swelling (ROR = 13.94, 95% CI: 3.93-49.4). For the second dose, the most reported signal was Injection site erythema, and the highest ROR was observed for Skin laceration (ROR = 14.67, 95% CI: 4.3–50.07). Among the 12 common signals, Injection site pruritus was associated with a higher risk after the second dose ($P = 7.76e\text{-}03$) (Figure 4(F), S3(B)). Among the unique signals, the most reported was Rash erythematous, and in the

Figure 4. (A) Forest plot of positive PT signals for MMRV vaccination in different age subgroups. (B) Curve plot depicting AEFI onset times for MMRV vaccines in different age subgroups. (C) Forest plot depicting positive PT signals for MMRV vaccination in different sex subgroups. (D) Comparison of unique PT signals between the male and female groups. (E) Forest plot depicting positive PT signals for MMRV vaccination in different dose subgroups. (F) Comparison of unique and differential PT signals between the first dose and second dose groups. G) Curve plot depicting AEFI onset times for MMRV vaccines in different dose subgroups. (H) Forest plot of positive PT signals for MMRV vaccination in different vaccination brand subgroups. (I) Comparison of unique and differential PT signals between the ProQuad and the other brands. (J) Curve plot depicting AEFI onset times for MMRV vaccines in different vaccination brand subgroups. PT, preferred term; ROR, reporting odds ratio; CI, confidence interval. The red dashed line indicates the reference line of ROR = 1, which represents no statistical association. The blue arrows indicate that the lower boundary of the 95% confidence interval for the ROR is higher than 50.

first dose and the same signal as the most frequent overall in the second dose. Statistically significant differences were observed in the onset times of AEFIs between the dose subgroups ($P < .0001$). The second dose resulted in AEFIs in 92.2% of cases within 7 d after vaccination, indicating an earlier onset compared with the first dose (Figure 4(G)).

Brand subgroup analysis

The VAERS database records two main types of MMRV vaccines produced by Merck & Co., ProQuad® and other brands. We compared the AEFIs between ProQuad and other MMRV vaccines, including 13,272 and 452 reports, respectively. The numbers of positive signals were 40 and 16 respectively (Figure 4(H)). In the ProQuad group, the most frequently reported symptoms were Injection site erythema, and the highest ROR was for Conjunctival disorder (ROR = 64.56, 95% CI: 6.71-620.66). A total of 60% of AEFIs were categorized under General disorders and administration site conditions and Skin and subcutaneous tissue disorders at the SOC level. In the other-brand group, the most frequently reported symptom was Generalized rash, with the highest ROR for Juvenile idiopathic arthritis (ROR = 37.18, 95% CI: 11.41-121.15). Only one signal, Hyperpyrexia, was common to both groups, with a lower in the ProQuad group ($P = 1.65e\text{-}4$). The specific signals numbered 39 and 15, respectively, consistent with the distribution of the most frequently reported signals within each group (Figure 4(I), S3(C)). Statistically significant differences were observed in the onset times of AEFIs between the two groups ($P = 1.42e\text{-}11$),with earlier onset in ProQuad group (Figure 4(J)). However, these findings should be interpreted with caution, as it may be influenced by reporting bias rather than reflecting a true difference in the vaccines' pharmacological properties.

Comparative analysis of MMRV and MMR and Varicella

We compared the AEFIs associated with MMRV vaccination and the separate administration of MMR and Varicella vaccines. Among the 62,511 reports for MMR vaccination, 11.9% involved severe cases. Of the 17 positive signals, the one most frequently reported were Rash maculo-papular, Feeling hot, and Autism (Figure S4(A)). with highest ROR observed for Mumps (ROR = 146.22, 95% CI: 72.56-294.64). Among the 47,467 Varicella vaccination cases, 6.4% were severe. Of the 28 positive signals, the most frequent were Infection, Dermatitis bullous, and Viral infection (Figure S4(B)). Overall, the AEFIs associated with MMRV were mostly mild skin-related symptoms and showed lower risk than those associated with MMR and Varicella. The onset times of AEFIs differed significantly among the three vaccine types (P <0.0001). The onset time for MMRV AEFIs was the earliest, with 86.5% occurring within 7 d after vaccination, whereas the Varicella vaccine showed the latest onset, with only 57.3% of AEFIs occurring within 7 d (Figure S4(C)).

Sensitivity analysis

A sensitivity analysis was conducted on the AEFIs of the MMRV vaccine (both overall and by subgroups) to assess the robustness of the main findings. After excluding the signals unrelated to adverse reactions, no additional signals were identified. Meanwhile, all the key signals discussed in the main text (such as seizures in epilepsy, wheezing in asthma, and conjunctival diseases) were still robustly detected under this standard.

Discussion

This study conducted a comprehensive pharmacovigilance analysis of the safety of MMRV vaccination in children under 7 y of age using VAERS data collected from 2005 to 2025. Multidimensional mining of 13,724 reports meeting the inclusion criteria systematically revealed distinct risk characteristics associated with MMRV vaccination in the pediatric population, particularly emphasizing the need for safety vigilance in children with underlying chronic diseases. These findings provide essential empirical evidence for optimizing immunization planning strategies.

By integrating four disproportionality algorithms (ROR, PRR, BCPNN, and MGPS) with the DME list specified by the EMA, we observed that the MMRV vaccine was generally safe for children in the overall population. Most AEFIs (94.1%) were non-severe and primarily manifested as local symptoms at the injection site such as erythema and swelling, consistent with the findings of Klein et al. in the VSD study.[12] Serious outcomes have become increasingly rare with the currently implemented immunization programs. Rigorous pre-marketing clinical trials, improvements in vaccine formulations, greater familiarity of healthcare providers with vaccines and enhanced adverse event management have collectively contributed to improve the safety of vaccines.[13]

The annual reporting data showed two distinct peaks. The first occurred in 2007 likely because many children received their initial vaccination in 2006–2007 following FDA approval of the vaccine for marketing.[14] During the early phase after introduction, medical institutions demonstrated high sensitivity to active surveillance and AEFI reporting. However, subsequent studies revealed that the risk of febrile convulsions within 5–12 d after the first MMRV vaccination was 2–4 times higher than that following separate MMR and varicella vaccinations,[15] which may have contributed to vaccine hesitancy, leading to a subsequent decline in vaccination rates and reports. As research on MMRV vaccine safety expanded, the Advisory Committee on Immunization Practices (ACIP) officially recommended a two-dose varicella vaccination schedule in 2010 (first dose at 12–15 months, second dose at 4–6 y). Between 2014 and 2024, the number of VAERS reports remained stable at approximately 900 cases per year (with a transient decrease during the COVID-19 pandemic period). Additionally, the launch of the VAERS online reporting platform in 2015 increased the willingness to report among medical institutions and the public, resulting in another peak in 2017.

A detailed analysis of AEFI onset times revealed distinct temporal evolution patterns. The Weibull distribution model shape parameter β clearly demonstrated the "early failure" characteristic : 46.0% of events occurred within 24 h after vaccination, 86.5% within one week, and 96.8% within 30 d. These findings strongly support the biological plausibility of acute AEFIs and highlight the importance of immediate post-vaccination observation, providing theoretical justification for the 30-d monitoring window recommended by disease control and management departments.

Key findings and clinical significance

An increase in the risk of conjunctival diseases and mental disorders

Overall, the risk signals of Conjunctival disorder and Oppositional defiant disorder were elevated, which may involve multiple factors including immunological, neuroinflammatory and genetic susceptibility mechanisms. One hypothesis suggests that viral antigens in the MMRV vaccine may share molecular similarities with conjunctival tissue antigens. Following vaccination, activated cross-reactive T cells could potentially attack conjunctival epithelial cells, leading to conjunctival congestion, follicular hyperplasia and increased secretions.[16] Similarly, regarding mental disorders, it has been proposed that the systemic immune response activated by the vaccine may lead to elevated levels of cytokines such as IL-1β and IL-6. However, these neuroinflammatory mechanisms remain speculative and are not yet supported by direct clinical evidence in the context of MMRV vaccination. Moreover, previous studies have not demonstrated a reliable association between MMRV vaccination and mental disorders.[17]

The risk of immune thrombocytopenia (ITP) should also not be underestimated

ITP was identified as a key risk after screening the DME list. The primary pathogenesis of ITP involves the immune system's loss of tolerance to self-antigens, resulting in enhanced immune-mediated platelet destruction and insufficient immune-mediated megakaryocyte platelet production. The pathogenesis of vaccine-related thrombocytopenia is thought to involve an immune-mediated response, potentially triggered by antigenic similarity between vaccine components and platelet antigens. This proposed mechanism raises a theoretical concern that subsequent exposure to the same antigen, through re-immunization, may pose a theoretical risk of disease recurrence. In clinical practice, this theoretical risk can contribute to heightened caution and vaccine hesitancy among both healthcare providers and parents. According to current statistics, vaccine-related thrombocytopenia is rare. The US VAERS recorded 177 cases of thrombocytopenia in children under seven y of age from 2005 to 2025, with MMRV accounting for 14.1% of reports. The most commonly associated with ITP was PNEUMO (PREVNAR13) (28.8%). In China, 773 cases of thrombocytopenia were reported from 2010 to 2015,[18] with an incidence rate of 0.20–0.35 cases per million doses. ITP-related AEFIs were primarily associated with the measles-mumps, IPV, and hepatitis B vaccines.

Targeted monitoring should be implemented for children with chronic diseases

To analyze special populations, we focused on three subgroups: epilepsy, asthma, and chronic kidney disease. These conditions are clinically eligible for vaccination during stable periods, but vaccination delays of 45–68%,[19] representing a gap in public health decision-making. Children with epilepsy have a significantly higher risk of Seizure (ROR = 10.27, case : control = 405 vs 17,176) after receiving MMRV. This risk estimate is higher than the relative risk of 1.98 for febrile seizures following MMRV (vs. separate MMR+V, case : control = 83,107 vs 376,354) in the general pediatric population reported by Klein et al.[12] This discrepancy is expected and underscores a critical distinction: our analysis specifically quantified the risk increment within a high-risk subgroup (children with preexisting epilepsy) compared to the general pediatric population, whereas Klein et al. compared the risk between two vaccine strategies in a broad, unselected cohort. This group may have an inherent susceptibility to seizures, which can be acutely triggered by febrile responses. This risk could be potentiated by the MMRV vaccine, as the varicella component has been suggested to enhance the immunogenicity of measles antigens, potentially leading to a more pronounced release of pro-inflammatory cytokines that may further lower the seizure threshold.[20] This risk is particularly pronounced within 48 h after vaccination (63.2%) and may also relate to the ability of attenuated live viruses to cross the blood-brain barrier and increase neural excitability.[20] Age is another critical factor compared with children aged 12–18 months, those aged 4–5 y may have a more mature immune system, which potentially produces a stronger inflammatory response to the vaccine and results in more reported cases (an increase of 57.1%). Additionally, fever may alter sodium channel function and induce mitochondrial stress, triggering abnormal neuronal discharge.[21] Notably, the five unique neuro-specific signals identified in the epilepsy subgroup indicate that clinicians should enhance neurological monitoring in these children. In children with asthma, specific respiratory system signals such as Wheezing and Respiratory tract congestion were identified, generally occurring within 24 h. The acute onset of these events suggests a potential triggering effect of the vaccine in susceptible individuals. The underlying mechanism may involve an enhanced inflammatory response in the airways, potentially exacerbated in children with preexisting asthma or atopic predisposition.[22] These findings highlight the importance of monitoring for early respiratory symptoms following MMRV vaccination in this subpopulation. It is important to note that children with epilepsy or asthma already have a higher potential risk of experiencing epileptic seizures or asthma symptoms. Therefore, the signals found in these subgroups should not be interpreted as the vaccines “causing” these conditions directly, but rather as the vaccines possibly triggering or exacerbating acute events in individuals with biological susceptibility. The observed temporal changes provide important support for this triggering hypothesis. Children with chronic kidney disease did not show a significantly elevated risk. The result is preliminary due to the relatively small sample size in this subgroup while they support the safety of vaccination in this group, consistent with current guidelines, further studies are needed for confirmation.

Refined subgroup analysis revealed the differences in risk

The rate of severe cases in the 1–1.5-year-old vaccination group was 10.8%(95% CI: 9.9%-11.8%), which was higher than that in other age groups. Penile swelling, unique to the first dose, contrasted sharply with injection site pruritus, which was more common after the second dose ($P = 2.56e\text{-}13$). Penile swelling may represent a localized inflammatory or hypersensitivity response,[23] which is generally self-limiting according to clinical observations. However, it is important to differentiate this reaction from other conditions such as urinary tract infection or trauma. The presence of systemic symptoms, such as breathing difficulties, should prompt evaluation for a more serious allergic reaction. Sex-based analysis showed that females were more prone to thrombotic thrombocytopenic purpura and Formication (ROR = 20.49) whereas males had Pain in extremity as the main unique signal. These findings suggest the need to pay attention to sex-based dimorphisms in immune responses. Vaccine brand analysis found that the ProQuad vaccine had a lower risk of hyperpyrexia ($P = 1.65e\text{-}4$) whereas the ROR of Juvenile idiopathic arthritis was high in other vaccine brands. Similarly, the differences in the onset time of AEFIs observed among different brands, although statistically significant ($P = 1.42e\text{-}11$), may not represent clinically or biologically significant differences. Factors such as the dominance of a certain brand (ProQuad®) in the dataset and possible variations in reporting behavior could also be at play.

No significant AEFIs differences were found between MMRV and separate MMR + Varicella vaccinations

Casabona et al.[24] reported that the relative risk of febrile convulsions increased approximately twofold within 5–12 d after the first dose of MMRV, whereas no increase in relative risk was observed in children aged 4–6 y. Most other studies have confirmed[14,25] that the MMRV vaccine and the separate MMR + V vaccines demonstrate comparable seroconversion rates and show no significant differences in AEFIs. Moreover, the AEFIs rate of MMRV vaccine is relatively low, predominantly presenting as local skin reactions.[26] Notably, a high-risk signal for Autism was detected in MMR vaccine AEFIs, compared with 57.3% for Varicella and 73.8% for MMR during the same period. These results provide a reference for developing differential monitoring strategies following vaccination.

Limitations

VAERS is subject to several inherent biases including selection bias (under-reporting and differential reporting by healthcare providers and patients), information bias (variability in the completeness and accuracy of reports), and confounding by factors not captured in the database. While our study employed multiple disproportionality algorithms and subsequently controlled the false discovery rate to mitigate the risk of false positives, the passive surveillance feature of the VAERS system entails several limitations. The data include incomplete, inaccurate, coincidental or unverified information, and it does not obtain follow-up records for each report, which compromises data quality and completeness. Furthermore, VAERS did not establish a control group of unvaccinated individuals, so we are unable to clearly distinguish the relationship between vaccine-related risks and the risks associated with the underlying diseases themselves. Critically, VAERS does not verify causality for reported events, caution should be exercised in interpreting the research results. Therefore, while the ROR and other disproportionality measures can quantify reporting associations, the signals identified in this study do not imply causation and should be interpreted as statistical hypotheses requiring further validation. Future work should prioritize prospective studies through active surveillance systems and strengthen experimental research into immune response mechanisms in children with underlying medical conditions.

Implications for public health policy

These findings hold dual significance for alleviating vaccine hesitancy. On one hand, providing comprehensive safety data can enhance parental confidence in childhood immunization. On the other hand, developing precise risk management strategies for special populations can increase vaccination coverage among high-risk groups. Specifically, we recommend the following actionable measures: First, for children with epilepsy, consider the preventive use of antipyretics around the time of vaccination, coupled with pre-vaccination counseling to address specific neurological concerns. Second, for children with asthma, vaccination should be avoided during acute phases, and pre-vaccination health reviews are advised to confirm respiratory stability. Third, it is advisable to incorporate a post-vaccination observation window of 7 d into practice guidelines, enabling timely detection and management of potential adverse events such as fever or rash. Notably, the comparative analysis of MMRV and separate MMR and varicella vaccinations shows that the combined vaccine may not introduce additional risks, which could help streamline immunization schedules and improve coverage rates.

Conclusion

This study conducted a comprehensive real-world pharmacovigilance analysis of AEFIs following MMRV vaccination in children under seven y of age using VAERS data, with a particular focus on those with underlying chronic conditions such as asthma, epilepsy, and kidney disease. The findings suggest a generally favorable safety profile for MMRV vaccination based on disproportional reporting analysis, though causal relationships cannot be established. Notably, 94.1% of the reported AEFIs were non-serious. The analysis indicated an increased reporting of seizures in children with epilepsy and a 6.81-fold higher reporting rate of wheezing in those with asthma. Variation in AEFI reporting were also observed across age, gender, dose, and vaccine brand. Temporal analysis using the Weibull model suggested that most AEFIs occurred early after vaccination. Overall, this study provides valuable risk-characterization information supporting tailored monitoring and preventive strategies for children with specific health conditions.

The signals identified in this study, particularly those in high-risk subgroups, should be interpreted as hypothesis-generating. Future research should aim to validate these risk patterns in more robust datasets. A critical next step would be to link these findings with active surveillance systems, such as the VSD or the Post-licensure Rapid Immunization Safety Monitoring (PRISM) system. These systems, with their well-defined cohorts and ability to calculate incidence rates, can confirm and quantify the potential risks identified here, moving from signal detection toward causal inference and ultimately informing more precise clinical guidelines.

Author contributions

CRediT: **Kai Yan:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing; **Guang Yang:** Resources, Supervision, Validation; **Lixuan Yan:** Investigation, Software, Visualization; **Ruopei Wu:** Data curation, Formal analysis; **Li Wu:** Data curation; **Yang Wang:** Methodology, Resources; **Peifeng He:** Conceptualization; **Qi Yu:** Funding acquisition, Supervision.

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Data availability statement

In this study, all data were obtained from publicly available databases and references to the available data can be accessed at https://vaers.hhs.gov/data. However, the identities of the medical reviewers are confidential and protected.

Ethics approval and consent to participate

This study conducted a secondary analysis of reported data from the public VAERS database and did not require ethical approval.

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